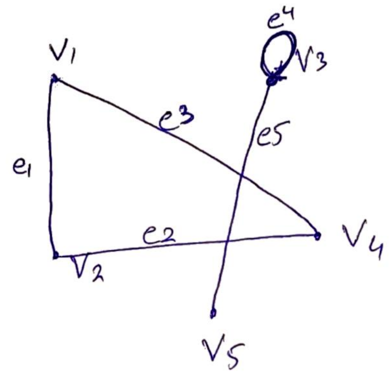
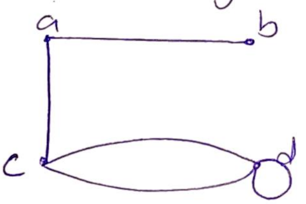
Q1:



Q2:

1. Total degree of graph $= 4+3+2+1$
$= 10$ (even)

There are many solutions.



deg(a) = 2
deg(b) = 1
deg(c) = 3
deg(d) = 4

2.

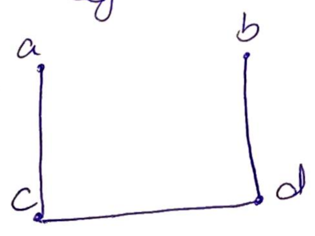
No of edges $= (1+1+3+3)/2$
$= 8/2 = 4$

so it will need to have 4 edges according to degree, so graph not possible.

3.

Total degree of G $= 1+1+2+2$
$= 6$ (even)

~~many~~



many solutions possible

deg(c) = deg(d) = 2

deg(a) = 1
deg(b) = 1

Q3

Fig 1.

deg(v1) = 2 = deg(v3)

deg(v2) = 4 = deg(v4) = deg(v5)

As every vertex has even degree so it is possible

Euler Circuit:

$v_1 e_1 v_2 e_2 v_5 e_3 v_2 e_4 v_3 e_5 v_4 e_6 v_5 e_7 v_4 e_8 v_1$

Fig 2.

deg(v1) = 5

v1, v7, v8, v9 has odd degree

so graph can't have Euler circuit

Q4:

1. Walk
2. Path
3. Simple Path
4. Simple Circuit
5. Circuit

Q5

(a) deg (a) = 4 = deg(b) = deg(c) = deg(d)

deg (e) = 6

all even so Euler Circuit exist.

a e c d e a e b d c b a

(b) deg(f) = 3 = deg(c)

so Euler Circuit don't exist

c d a e c b d e f b a f

Q6: for Hamiltonian Circuit

→ H contain every vertex of G

→ H is connected

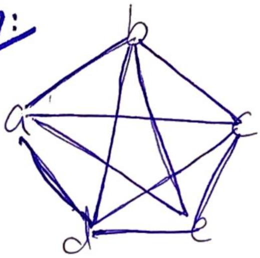
→ H has same no. of edges as vertices

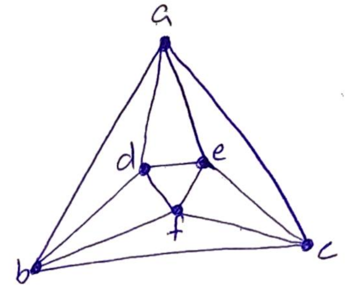
→ Every vertex of H has degree 2.

Fig 1.

deg(c) = 3 = deg(f)

Q7:





4 - Regular Graph

Q8:

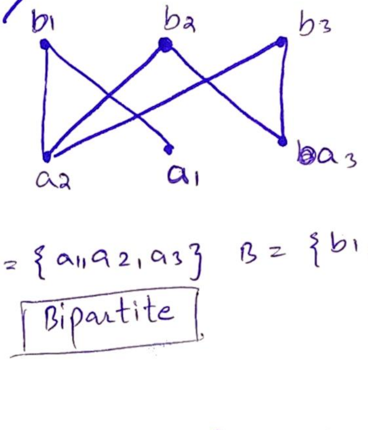
Fig 1. Connected

Fig 2. Connected

Fig 3. Not connected

Q9

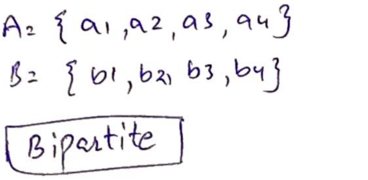
1.



A = {a1, a2, a3} B = {b1, b2, b3}

Bipartite

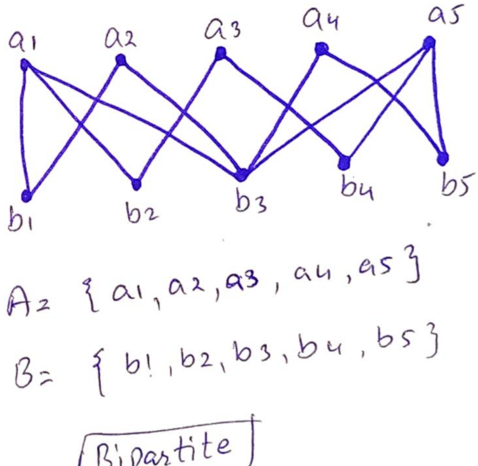
2.



A = {a1, a2, a3, a4}
B = {b1, b2, b3, b4}

Bipartite

3.



A = {a1, a2, a3, a4, a5}

B = {b1, b2, b3, b4, b5}

Bipartite

Q10.

1.

	a	b	c	d
a	0	1	1	1
b	1	0	0	1
c	1	0	0	1
d	1	1	1	0

2.

	a	b	c	d	e
a	0	1	0	1	0
b	1	0	0	1	1
c	0	0	0	1	1
d	1	1	1	0	0
e	0	1	1	0	0

3.

	a	b	c	d
a	1	1	1	1
b	0	0	0	1
c	1	1	0	0
d	0	1	1	1

4.

	a	b	c	d	e
a	0	1	0	1	0
b	1	0	1	1	1
c	0	1	1	0	0
d	1	0	0	0	1
e	0	0	1	0	1